



System Subsystem Design Description
for the
The Balance Project
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DOCUMENT CHANGE HISTORY

The following table is a simple list of released revisions sent for review. Records of reviews and the review artifacts are saved with reviewer information in the The Balance Project artifact repository.

Change Record

Date	Version	Author(s)	Change Reference
13 Sep 2025	P1	Vinay Agarwal	Preliminary DRAFT version
5 Dec 2025	P1	Vinay Agarwal	Final Draft of the system

Draft P1 Preliminary version of this document.

1. Baseline Document draft 1
2. Final fix and update from the first draft



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CHAPTER 1

Scope

This document provides the System / Subsystem Design Description (**SSDD**) for the Balance System. The system will be referred to as the Balance System.

1.1 Identification

The Balance System described in this document shall be known as Balance System version 1. However, the Operational Concept Description **OCD** described herein shall be applicable to pre-releases such as Beta-releases for a phased release as listed for each requirement. The major system interfaces and capabilities are fully specified in Chapter 3.

1.2 System Overview

Figure 1 shows the high-level architecture for the Balance System system. This diagram shows the major external interfaces that provide the capabilities of Balance System.

The Balance System is a single player game. The game is a standalone game played on the STM32 board. This system would be a game where the user would have to balance a ball on a LCD screen that is builtin on the STM32 board. The objective of the game is to balance the ball on the screen based on the way the board was tilted. Using an onboard gyroscope that communicates across the board to the MCU though SPI communication. Balance System would keep track of the current position of the ball and where the next updated move is. This helps keep track of the system of where the ball is until a movement has occurred. Balance System shall process at a maximum 180 Hz. This would give the user enough time to process the current angle of the ball and be able to present on the LCD-TFT screen.

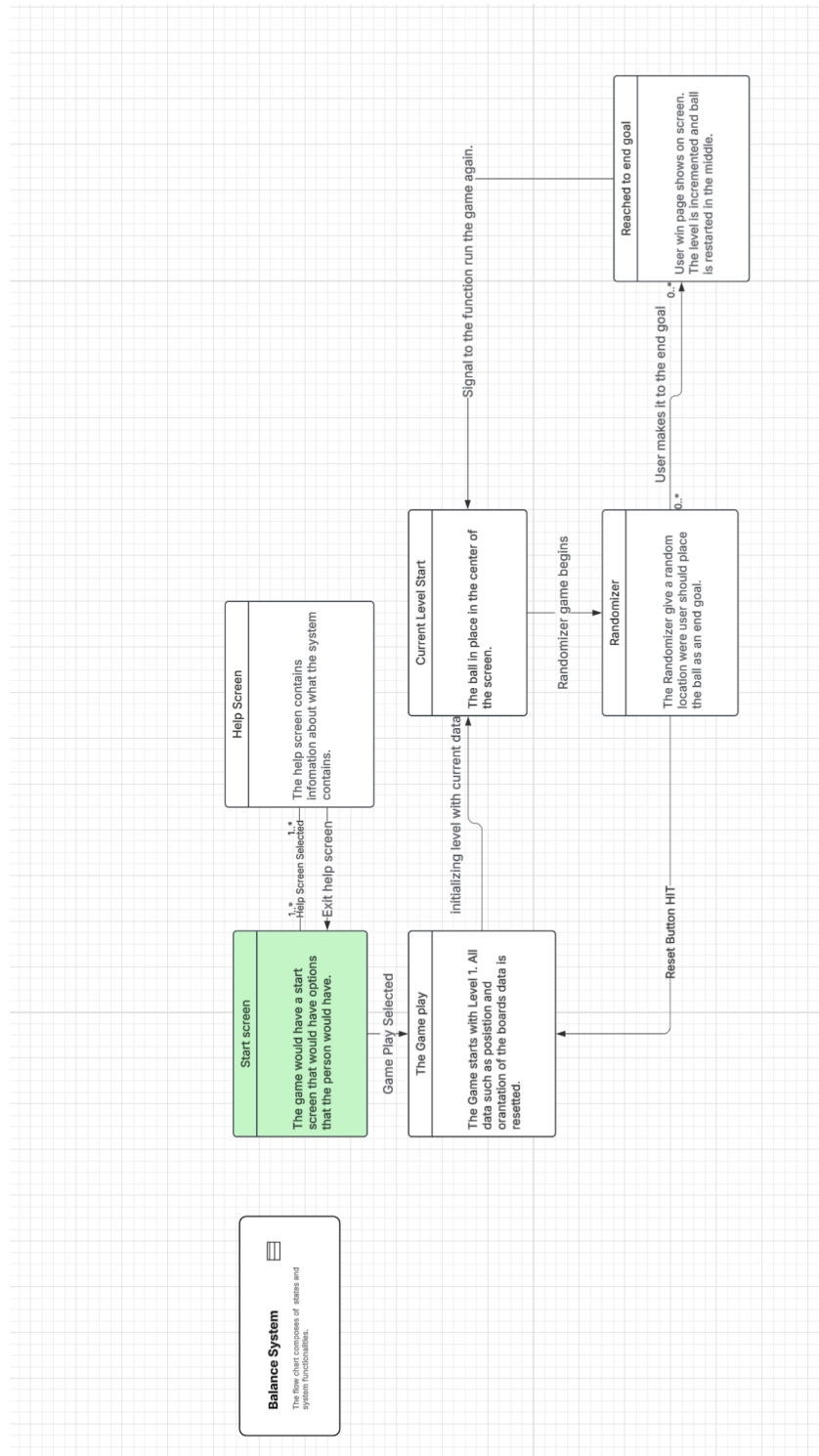


Figure 1: System Overview



1.3 Document Overview

This section provides information about this document's security/privacy considerations, contents, structure, and version information. This section also provides information regarding how specifications are formatted in this artifact and how they can best be understood.



CHAPTER 2

References

This section provides a list of referenced items for this document.

2.1 Acronyms and Abbreviations

This section defines acronyms and abbreviations used in this and related documents.

Table 1: Acronym Definitions

Acronym	Definition
YSM	Yourdon Structured Method
LTDC	Layered Transmission Display Controller
STM	STMicroelectronics
MEMS	Micro-Electro-Mechanical Systems
End of acronym definition table	

2.2 Glossary and Definitions

This section defines glossary terms used in this and related documents.

Table 2: Glossary Terms and Definitions

Glossary Term	Definition
STM32F429I	Micro-controller board has all component fit onto one board.
Customer	The professor that is view the grading all assignments.
End of glossary terms table	

2.3 Referenced Documents

This section lists the referenced documents for this document. The references are categorized into two categories:



External Documents not directly associated with this project.

Project Documents that are directly associated with this project.

2.3.1 External Documents

2.3.2 Project Specific Documents



CHAPTER 3

System-wide Design Decisions

The system will always start at the initial state. The initial state would run through all initialize steps to do reading for the system. That means that the Gyrocode will state doing reading and the data structure will be stored in memory. Once the initialization begins then the user should be able to view the main screen that is used for the system. This Main screen is where the user can select 2 buttons on the screen. The user would have 2 options to be able to select the buttons on the main screen. One being touch screen and the other with a scroll button. Both options are located on the board. The Main screen is where the user would select to start the game or read the helpful guide. If the user select the help screen, then a new page would appear. This new page will descriptions of how to play the game. The page will have symbol descriptions and what the users' should be looking for while playing the game. To return to the main screen the users should click the 'X' on the top right. If the user selects the run game button, the game would begin. When the game begins from the main page it always starts at level one. Reach levels page would have the same setup. This setup allows the user how much time they have left at the level and the level there are on. This allows the user to see what then current state of the level they are play that they are at. Every single level that the user is playing the game the ball is place in a random location. The goal where the user should place the ball whould also be placed in a random location. This allows every level to be unique. If the user does not meet the goal at the end of the round then then user losses the game and is taken back to main menu. If the user moves the ball into the goal line then the level is complete and the game is continued.



CHAPTER 4

System Architectural Design

This chapter describes the system architectural design.

4.1 System Components

The System components compose of the following: GYRO scope : reads angular velocities LTDC : screen and touch screen to display angles. SDRAM : hold memory displaying and updating the screen.

4.2 Concept of Execution

First reads the the gyroscope value. Calculates what angles the ball should rotate to. Determain if there is match with the posistion of the ball and the goal.

4.3 Interface Design

3 external interfaces. Gyroscope, SDRAM, and LTDC screen. Then are all interfacing with the STM32ZIT chip on the board.

4.3.1 Interface One

The Gyroscope Interface. The gyroscope in face is an L3GD20. This will be our main source of gyroscopic information that is used to understand accelerations based off a role pitch and y'all we will be only using roll and pitch for this project.

4.3.2 Interface Two

SDRAM Interface An STM32F49i likely refers to the STM32F429 or a similar high-performance F4 series MCU, which offers internal SRAM (e.g., 256KB on some F429s) and an extensive Flexible Memory Controller (FMC) to interface with external RAM like SDRAM, PSRAM, or NOR/NAND Flash for massive memory expansion, crucial for graphics and complex applications

4.3.3 Interface Three

LTDC Interface This scenario describes how to configure the STM32 LTDC display controller and TouchGFX Generator when using a LCD-TFT parallel display interface.



CHAPTER 5

Traceability

The systems traceability will be in the order of OCD to SDP to SSS. Each requirement would be traced back from one to the other based on there numbering.



CHAPTER 6

Schedule

The Schedule is based on the time line of the professor.



APPENDIX

Notes

This section provides notes, as necessary, to document the system /subsystem design description.



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